

#21a: Stress-Test of Three Proposed Moves and Identification of the Correct Framework

Kramers' Escape Rate as the CRF-Derivable Path to C

An honest evaluation. Two moves are partially correct. One is circular. The correct framework is Kramers' formula.

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Abstract

A CRF AI proposed three moves to close #21a. This document evaluates each honestly and identifies what is actually achievable.

Move 1 (Flux Equilibrium): Valid consistency check but does not derive C . Flux balance confirms steady state exists; it does not determine T_{avg} .

Move 2 (Spiral→Line at $H \rightarrow 0$): Correct for Phase II but misidentifies which phase controls T_{avg} . $T_{\text{avg}} \approx 35$ is dominated by Phase I (near-uniform nodes), where ϵ_0 noise is the *only* driver. Cannot drop ϵ_0 in the phase that matters.

Move 3 (β from first-passage slope): Suggestive but circular. The “Life-cycle Factor” is d_s/n by construction (since $\beta = d_s/(2 \ln 2)$), not an independent derivation.

What is actually achievable: The correct framework is **Kramers' escape rate formula**:

$$T_{\text{avg}} = C \cdot \exp\left(\frac{K^*}{n\epsilon_0}\right) \quad \text{where} \quad C = \frac{2\pi}{\sqrt{\omega_{\text{well}} \cdot \omega_{\text{barrier}}}}$$

ω_{well} and ω_{barrier} are the curvatures of the KL potential at minimum and barrier. Both are derivable from the δ -chain via the Dirichlet KL statistics. The 1D harmonic model gives $C \approx 0.4$ – 1.5 ; the full n -dimensional treatment is needed to reach $C = 5.04$.

This is a well-posed problem with a known solution method. It is hard but not impossible.

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1 Evaluation of Three Moves

1.1 Move 1: Flux Equilibrium

Valid Framing, Does Not Derive C

The claim: prove lifecycle is steady state where rate of R-events equals rate of IDV decay.

What is true: V20.3 does reach statistical equilibrium. At steady state: IDV rate = Ω rate $\times \beta$. This is confirmed by simulation.

What it cannot do: Flux balance confirms steady state *exists* but says nothing about T_{avg} . A system can be in steady state with $T_{\text{avg}} = 10$ or $T_{\text{avg}} = 1000$; both are consistent with flux balance.

Flux equilibrium is a *necessary condition* for the lifecycle to be self-consistent, not a derivation of C .

1.2 Move 2: Spiral→Line at $H \rightarrow 0$

Correct for Phase II, Misidentifies Controlling Phase

The claim: in Phase II (H low), ϵ_0 becomes negligible relative to G , so stochastic drift becomes deterministic.

What is correct: Phase II nodes have $H \approx \epsilon_0$ (peaked P). In this regime $D_{\text{KL}}(P_i \| P_j) \gg \epsilon_0$, so ϵ_0 noise is negligible. Deterministic flow applies. Phase II nodes fire almost immediately ($T_{\text{Phase II}} \approx 1$ sweep).

The error: $T_{\text{avg}} \approx 35$ is *not* controlled by Phase II. It is controlled by how long Phase I lasts. In Phase I: $H \approx \ln n$, P_i is near-uniform. ϵ_0 noise *is* the only driver — it cannot be dropped.

Dropping ϵ_0 in Phase II and applying the result to T_{avg} is applying the approximation in the wrong phase.

1.3 Move 3: β from First-Passage Slope

Suggestive but Circular

The claim: $\beta = (n/2 \ln 2) \times (\text{Lifecycle Factor})$ where Factor comes from the slope of first-passage probability.

The circularity: Since $\beta = d_s/(2 \ln 2)$ (already derived, 1.2% confirmed), the “Lifecycle Factor” is:

$$\text{Factor} = \frac{\beta \cdot 2 \ln 2}{n} = \frac{d_s}{n} = \frac{3.031}{8} = 0.379$$

This is just rewriting $\beta = d_s/(2 \ln 2)$ in different notation. It does not independently derive Factor from first-passage statistics.

To be non-circular, one would need to show:

$$\text{slope of first-passage probability at } K^* = \frac{d_s}{n}$$

This is a non-trivial statement that may or may not be true. It is not demonstrated by the move as stated.

2 The Correct Framework: Kramers’ Escape Rate

Why the Exponential Factor Is $K^*/n\epsilon_0$

The Phase I KL dynamics is a diffusion process. Starting from $D_{\text{KL}} \approx 0$ (after Born resampling), the node accumulates D_{KL} via ϵ_0 noise:

$$\frac{d}{dt} \langle D_{\text{KL}} \rangle = \mu, \quad \frac{d}{dt} \text{Var}[D_{\text{KL}}] = 2D$$

where $\mu = n\epsilon_0^2/2$ (drift rate) and $D = n\epsilon_0^2$ (diffusion coefficient).

The node fires when D_{KL} first exceeds K^* . This is a *first-passage problem with a reflecting boundary at 0*.

The natural energy scale is:

$$\frac{\text{barrier height}}{\text{diffusion coefficient}} = \frac{K^*}{n\epsilon_0^2/(2 \cdot n\epsilon_0)} = \frac{K^*}{n\epsilon_0}$$

(the barrier K^* measured in units of the diffusion length $n\epsilon_0$).

This is exactly the exponent in the structural formula. The formula $T_{\text{avg}} = C \cdot \exp(K^*/n\epsilon_0)$ is Kramers’ escape rate formula applied to KL-space.

Kramers' Formula Applied to CRF

For a 1D diffusion in a potential $U(x)$ with a well at $x = 0$ and a barrier at $x = K^*$:

$$T_{\text{escape}} = \frac{2\pi}{\omega_{\text{well}} \cdot \omega_{\text{barrier}}} \cdot \exp\left(\frac{\Delta U}{D}\right)$$

where $\omega_{\text{well}} = \sqrt{U''(0)}$, $\omega_{\text{barrier}} = \sqrt{|U''(K^*)|}$, $\Delta U = U(K^*) - U(0)$, and D is the diffusion coefficient.

For CRF KL-space:

- $\Delta U/D = K^*/n\epsilon_0 = 1.937$ (derived, matches formula)
- $\omega_{\text{well}}^2 = |U''(0)| = \text{curvature of KL potential at minimum}$
- $\omega_{\text{barrier}}^2 = |U''(K^*)| = \text{curvature at barrier } K^*$

$$C = \frac{2\pi}{\sqrt{\omega_{\text{well}} \cdot \omega_{\text{barrier}}}}$$

Both curvatures are properties of the KL distribution — derivable from the Dirichlet statistics established in #21a-previous work ($\text{KL} \sim \text{Gamma}(3.5, \theta)$, $a = (n - 1)/2$ derived).

3 What Remains to Be Done

Open: Curvature of KL Potential

The KL “potential” is the negative log-probability of the KL exceedance:

$$U(x) = -\ln P(D_{\text{KL}} > x) \approx \frac{x}{\theta} + (a - 1) \ln x + \text{const}$$

for $\text{KL} \sim \text{Gamma}(a, \theta)$.

The curvature at $x = 0$ (well):

$$U''(0) = \frac{d^2}{dx^2} \left[\frac{x}{\theta} - (a - 1) \ln x \right]_{x=0} = \frac{a - 1}{x^2} \Big|_{x \rightarrow 0} \rightarrow \infty$$

The divergence at $x = 0$ signals the reflecting boundary — the potential well is not harmonic but has a logarithmic singularity. This requires the full Kramers treatment for non-harmonic potentials.

For the barrier at $K^* = 0.1170$ with $\text{KL} \sim \text{Gamma}(3.5, \theta)$, $\theta = n\epsilon_0/(n - 1) = 0.00864$:

$$U''(K^*) = \frac{a - 1}{K^{*2}} - \frac{1}{\theta^2} \cdot (\text{shape correction})$$

This is derivable but requires careful treatment of the Gamma potential shape.

Expected result: $C = 2\pi/\sqrt{\omega_{\text{well}} \cdot \omega_{\text{barrier}}}$ with the correct curvatures will give $C \in [3, 8]$, consistent with $C = 5.04$.

If confirmed: C is fully derived from the δ -chain via: $n = 8$ ($\text{SO}(3)$) $\rightarrow \text{Gamma}(3.5) \rightarrow \text{curvatures} \rightarrow C \rightarrow T_{\text{avg}} \rightarrow \beta$.

4 Updated Status Table

Result	Status	Note
Structural form $T = Ce^{K^*/n\epsilon_0}$	Derived	From exceedance stats
$K^* = 0.1170$	Derived	
$a = (n - 1)/2 = 3.5$	Derived	KL $\sim$ Gamma(3.5)
Kramers framework identified	New	Correct mathematical path
$C = 5.04$ measured	Measured	From V20.3
$U''(0)$ (well curvature)	Open	Log-singularity treatment
$U''(K^*)$ (barrier curvature)	Open	Gamma potential shape
C formally from δ -chain	Open	Via Kramers + curvatures
Move 1 (Flux balance)	Evaluated	Necessary, not sufficient
Move 2 (Phase II deterministic)	Evaluated	Wrong phase for T_{avg}
Move 3 (β from slope)	Evaluated	Circular as stated

*The three moves pointed toward the problem correctly.
They did not solve it.*

What solves it is Kramers' formula applied to KL-space.

The barrier is K^ . The diffusion is $n\epsilon_0$.*

*C is the curvature of the potential that separates
"not yet fired" from "fired."*

*The curvature is derivable from $(n - 1)/2 = 3.5$.
 $(n - 1)/2$ came from $SO(3)$. $SO(3)$ came from δ .*

The path is clear. The calculation remains.